10/10

CHE 313 Transport Phenomena III

Quiz #3 (11/11/25) **Name:** Lucia Hernandez

$F = 10$ kmol $\quad x_F = 0.8$

10.0 kmoles of a feed that is 0.80 mole fraction acetone and 0.20 mole fraction ethanol is batch distilled in a system with a still pot (an equilibrium contact), a column that acts as one equilibrium stage (total two equilibrium contacts), a total condenser, and reflux to the column. p = 1.0 atm, the reflux ratio is L/D = 1.0, CMO is valid, and $x_{W,final}$= 0.20.

$x_f = 0.2$ $\quad x_{D}$ changing

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1) (2pts) Find the slope of operation equation, $y = (L/V)x + (1-L/V)x_D$

$\frac{L}{V} = \frac{L}{L+D} = \frac{L/D}{L/D+1} = \frac{1}{1+1} = \frac{1}{2}$

$y = \frac{1}{2}x + (1-\frac{1}{2})x_D$

$y = 0.5x + 0.5x_D$

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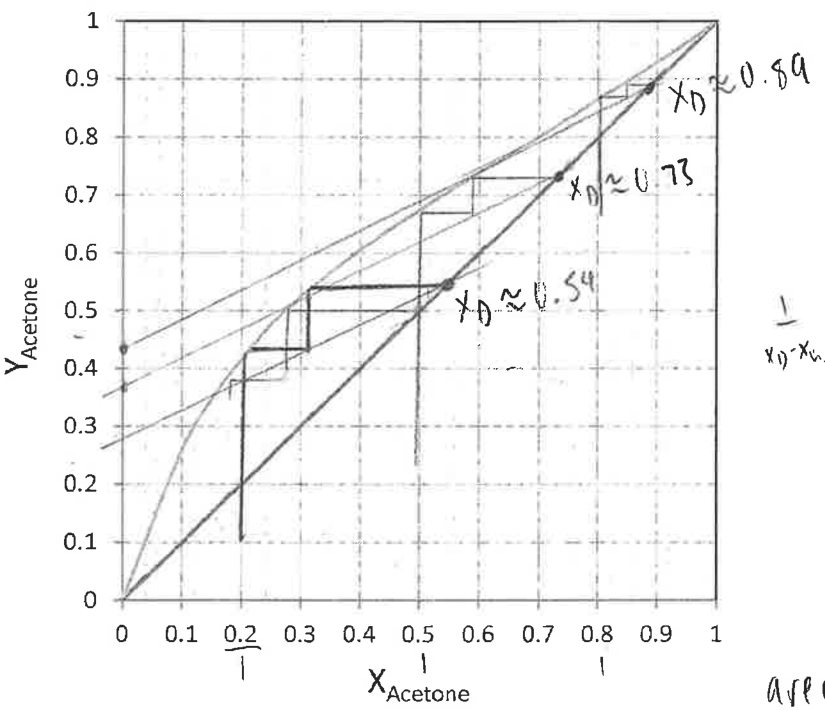
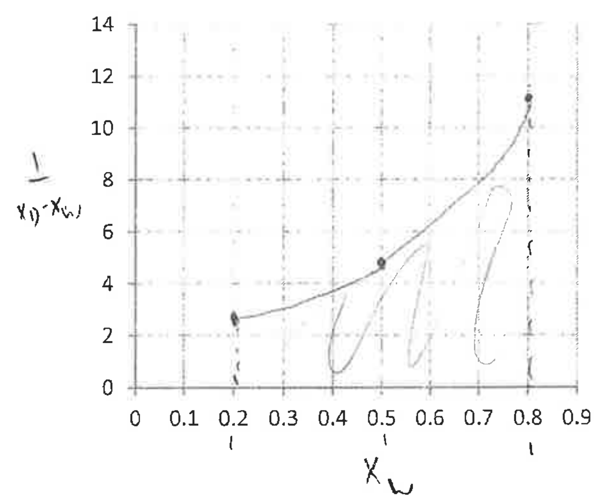
2) (4pts) Complete the table below using the VLE graph (draw OP lines and mark your x_w values on the graph), and mark points for area calculation.

$x_{mid} = \frac{x_F + x_f}{2} = 0.5$

	x_W	x_D	$x_D - x_W$	$1/(x_D - x_W)$
$x_{W,initial}$	0.8	0.89	0.09	11.11
$x_{W,middle}$	0.5	0.73	0.23	4.348
$x_{W,final}$	0.2	0.54	0.34	2.941





$area = \frac{0.8-0.2}{6}\left[2.941 + 4(4.348) + 11.11\right]$

$= 3.1443$

$D = F - W_f$

3) (4pts) Find W_{final}, D_{total}, and $x_{D,avg}$.

$W_f = Fe^{-area} = 10e^{-3.1443} = 0.431$ kmol

$D_{tot} = F - W_f = 10 - 0.431 = 9.569$ kmol

$x_{D,avg} = \frac{F x_F - W_f \cdot x_f}{D_{tot}} = \frac{(10)(0.8) - (0.431)(0.2)}{9.569} = 0.827$

Simpson's rule: $Area = \frac{x_{Feed}-x_{final}}{6}\left[\frac{1}{x_D-x_W}\Big|_{x_{final}} + \frac{4}{x_D-x_W}\Big|_{x_{middle}} + \frac{1}{x_D-x_W}\Big|_{x_{Feed}}\right]$

Rayleigh equation: $W_{final} = F\,exp\left(-\int_{x_{final}}^{x_{Feed}} \frac{dx_w}{(x_D-x_w)}\right) = F\,exp(-Area)$

$x_{D,avg} = (F*x_F - W_{final} * x_{final})/D_{total}$